

EVEN- p BOUNDS FOR NORM ATTAINMENT SETS ON ℓ_p^2

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SOURCE QUESTION

This packet concerns Debmalya Sain's paper *On the norm attainment set of a bounded linear operator*, arXiv:1608.00755 [1]. The paper proves that if $X = \ell_p^2$, $p \in \mathbb{N} \setminus \{1\}$, and $T \in \mathcal{L}(X)$ is not a scalar multiple of an isometry, then

$$|M_T| \leq 2(8p - 5).$$

Immediately after that theorem, the paper asks for the optimal bound.

$$|M_T| \leq 2(8p - 5) \iff |M_T| \leq 2(8p - 5).$$

□

Remark 2.2. We do not expect the upper bound $2(8p - 5)$ obtained in Theorem 2.7 to be an optimal upper bound for $|M_T|$. In fact, it seems to us that using stronger geometric and algebraic techniques, it might be possible to improve upon the bound $2(8p - 5)$. In view of this, we pose the following open question :

Open Question : For the Banach space $\mathbb{X} = \ell_p^2$, find an optimal bound for $|M_T|$, where $T \in \mathcal{L}(\mathbb{X})$ is not a scalar multiple of an isometry.

Remark 2.3. The case $p = 2$ deserves special mention, since in this case $\mathbb{X}$ is the two dimensional Euclidean space. It follows from Theorem 2.2 of [7] that if $T \in \mathcal{L}(\ell_2^2)$ then either $M_T = S_{\mathbb{X}}$ or M_T is a doubleton. In the first case, T is a scalar multiple of an isometry and in the second case T is smooth.

Transcription of the relevant question on PDF page 8:

Open Question: For the Banach space $X = \ell_p^2$, find an optimal bound for $|M_T|$, where $T \in \mathcal{L}(X)$ is not a scalar multiple of an isometry.

RESULT

We do not solve the full problem for all p . We prove a substantial algebraic improvement for all even p , and this completely settles the first non-Hilbert even case.

Theorem 1. *Let $p \geq 4$ be an even integer and let $T \in \mathcal{L}(\ell_p^2)$ be a real linear operator which is not a scalar multiple of an isometry. Then*

$$|M_T| \leq p.$$

Corollary 1. *For $X = \ell_4^2$, the optimal bound in Sain's question is 4.*

PROOF INTUITION

For even p , the p -th power of the norm is a polynomial, not merely a piecewise polynomial. After normalizing by $\|T\|$, the inequality $\|Tz\| \leq \|T\| \|z\|$ becomes a non-negative homogeneous binary form

$$P(z) = \|T\|^p \|z\|_p^p - \|Tz\|_p^p.$$

The norm-attaining directions are exactly the real projective zero lines of this form. A non-negative real polynomial cannot cross sign at a real zero line, so each such line occurs with even multiplicity. Since P has degree p and is not identically zero unless T is a scalar multiple of an isometry, there can be at most $p/2$ zero lines, hence at most p unit vectors in M_T . In degree 4, the usual Hadamard operator attains this bound.

PROOF OF THEOREM 1

Let T have matrix

$$T(x, y) = (ax + by, cx + dy)$$

with respect to the standard basis. Since p is even,

$$\|(x, y)\|_p^p = x^p + y^p, \quad \|T(x, y)\|_p^p = (ax + by)^p + (cx + dy)^p.$$

Put

$$P(x, y) = \|T\|^p (x^p + y^p) - (ax + by)^p - (cx + dy)^p.$$

The operator norm inequality gives $P(x, y) \geq 0$ for every $(x, y) \in \mathbb{R}^2$. Also, P is a homogeneous binary form of degree p .

If $P \equiv 0$, then

$$\|Tz\|_p = \|T\| \|z\|_p \quad (z \in \mathbb{R}^2).$$

Since the zero operator is a scalar multiple of any isometry, the hypothesis implies $T \neq 0$. Therefore $T/\|T\|$ is an isometry, contradicting the hypothesis. Hence P is not the zero polynomial.

Each projective norming direction $[x : y] \in \mathbb{P}^1(\mathbb{R})$ is a real line on which P vanishes. Conversely, if P vanishes on a non-zero vector, then equality holds in the operator norm inequality on that line, so the two unit vectors on that line belong to M_T . Thus $|M_T| = 2r$, where r is the number of distinct real projective zero lines of P .

We claim that every real zero line of P occurs with even multiplicity. Indeed, let L be such a line. Since P is a binary form, the real linear equation defining L divides P . Choose a point of L not lying on any other zero line, and use local coordinates (s, u) in which $L = \{u = 0\}$. If the multiplicity of L were odd, then locally

$$P(s, u) = u^m Q(s, u), \quad Q(s_0, 0) \neq 0,$$

with m odd, so P would change sign as u crosses 0. This is impossible because $P \geq 0$ everywhere. Therefore each of the r real zero lines contributes multiplicity at least 2. Since the total degree is p , we have $2r \leq p$. As $|M_T| = 2r$, this proves

$$|M_T| \leq p.$$

SHARPNESS FOR $p = 4$

Let $H : \ell_4^2 \rightarrow \ell_4^2$ be

$$H(x, y) = (x + y, x - y).$$

This operator is not a scalar multiple of an isometry, since

$$\|H(1, 0)\|_4^4 = 2 \quad \text{whereas} \quad \|H(2^{-1/4}, 2^{-1/4})\|_4^4 = 8.$$

On the unit sphere $x^4 + y^4 = 1$,

$$\|H(x, y)\|_4^4 = (x + y)^4 + (x - y)^4 = 2(x^4 + y^4) + 12x^2y^2 = 2 + 12x^2y^2.$$

Since $2x^2y^2 \leq x^4 + y^4 = 1$, the maximum is 8, and equality holds exactly when $x^2 = y^2$. On $S_{\ell_4^2}$, this means

$$(x, y) \in \left\{ \pm(2^{-1/4}, 2^{-1/4}), \pm(2^{-1/4}, -2^{-1/4}) \right\}.$$

Thus $|M_H| = 4$. Theorem 1 gives $|M_T| \leq 4$ for every non-isometric T on ℓ_4^2 , so the optimal bound is exactly 4.

VERIFICATION AND NOVELTY NOTES

The proof is purely algebraic. The accompanying script `code/verify_p4_example.py` numerically samples the ℓ_4^2 unit sphere for the Hadamard example and confirms that the only sampled maximizers are the four sign choices above. This is only a sanity check; it is not used in the proof.

Local cheap indexes were searched for arXiv:1608.00755, the phrases ℓ_p^2 , M_T , $|M_T|$, $8p - 5$, and norm-attainment-set variants. The closest local hit was the later paper arXiv:1806.01051, which improves Sain's numerical estimate only to $4(4p - 3)$ and does not give the even- p polynomial bound or the exact $p = 4$ answer. Web searches for the exact open question and for the bound $2(8p - 5)$ did not surface a later solution.

LIMITATIONS AND HUMAN REVIEW RECOMMENDATION

This packet does not settle odd p , and for even $p \geq 6$ it gives an upper bound p , not the exact optimum. Review should focus on the multiplicity argument for non-negative homogeneous binary forms and on the interpretation that each real projective zero line corresponds to exactly two points of M_T .

REFERENCES

- [1] D. Sain, *On the norm attainment set of a bounded linear operator*, J. Math. Anal. Appl. **457** (2018), 67–76; arXiv:1608.00755.
- [2] D. Sain, K. Paul, and K. Mandal, *Relation between norm attainment set and minimum norm attainment set of a bounded linear operator*, arXiv:1806.01051.